

AWS checklist and maintenance guide

pilot Jan
Pete Harkus

Station name	QAS U	
People present	JEB, MAV, Kenneth Hoag	instructed by:
Purpose	standard maintenance / installation / take-down / quick visit / emergency	
Date	22 Aug 2014	
Transport	helicopter / fixed wing / snow mobile / dog sleds / on foot from: 212	
Weather	warm / cold / very cold / no wind / some wind / rain / cloud cover %: 0	

Notes: no snow cover

Metadata before maintenance (if establishing a new AWS: skip this page)

Time difference between logger clock and UTC	logger 1:30 s past	+/-
Adjust logger time to UTC (be sure PC runs on UTC time)		☒ Y/N
Download data to PC and/or change CF Card (wait for green light)		Y/N
Name of logger program	2013c	
Photo or screen dump of values in fast scan mode		☒ Y/N
Latitude (dd mm.mmm)		
Longitude (dd mm.mmm)		
Photos of tripod, sensors, logger box wiring etc. (more is better)		☒ Y/N
Mast tilt in boom direction (+ if radiometer tilting down)	+2	°
Mast tilt across boom looking from radiometer (- if clockwise)	-10	°
Boom direction relative to north	170	° true/magn
Box on wind sensor exactly towards mast? If not, measure direction.		☒ Y/N
Radiometer aligned with mast and inclinometer? If not, measure.		☒ Y/N
Temperature / humidity height (bottom of casing – surface)	255	cm
Sonic ranger on AWS height (membrane – surface)		cm
Wind sensor height (center of propeller – surface)	293	cm
Sonic ranger on stakes height (membrane – surface)	286	cm
Free length of stake – outer, holding boom with sonic ranger	313	cm
Free length of stake – middle		cm
Free length of stake – outer, holding boom without sonic ranger	320	cm
Free length of ablation hose on/above the ice surface	14.95 m	(314) cm
Alt: depth of ablation hose using markings on hose		X cm
Height of ablation hose fluid level above the ice surface	14.95 m	cm
Vertical difference in surface height (at station – at hose) (approx)	153.25	cm
Length of thermistor string on surface from surface marking	465*	455** cm
Height of surface irregularities	max min 75 cm	locally smooth cm
Stake assembly still standing?	set screws worked	☒ Y/N / more or less
Apparent damage to sensors? If yes, which?	tensioner loose?	Y/N

JEB
MAN

precise
3 person
job

JEB 325

Notes:

6, 12 295

GPS 15 m from station @ 60° E of mag south

station summary taken

2013c

CF card complains it has not enough space
* to top of shrink tube, 455 cm to Therm 1

Therm 6 is 62 cm out of ice.

Part	Old part number	New part number, if replaced	Potential maintenance tasks
Radiometer		X	- Aligned with mast: Y/N - Clean: Y/N
Inclinometer		X	- Aligned with radiometer within 0.5°: Y/N
Satellite antenna		X	
Wind sensor		X	- Box on sensor exactly towards mast (south): Y/N
Temperature / humidity assembly		X	
T/hum assembly casing		X	- Fan spins and sounds OK: Y/N
T/hum assembly bracket		X	
Temperature probe in T/hum assembly		X	
HygroClip in T/hum assembly		X	- 40 °C offset in HygroClip temperature: Y/N
Sonic ranger on AWS		yes	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N
Solar panel		X	- Output OK: Y/N - Clean: Y/N
GPS antenna		X	- Stuck to top of logger enclosure: Y/N
Data logger		X	- New internal battery in old logger: Y/N - New operating system: Y/N
Card reader		X	- LED green of flashing orange (active): Y/N
Multiplexer		X	- Clicking as it should: Y/N
Iridium modem IMEI number		X	- Transmissions tested: Y/N - Problems: Y/N
Barometer		X	
Sonic ranger on stakes		yes	- Old sensor with new membrane: Y/N - Clicking 5 times as it should: Y/N - In danger of being buried next winter: Y/N
Thermistor string			- Old string left in ice: Y/N/cut
Ablation hose assembly		X	- Old hose left in ice: Y/N/cut - Air in hose: Y/N/removed - Bladder 50% full: Y/N / after emptying/filling
Logger enclosure	- Replaced (including everything inside): Y/N - New vent filter in logger enclosure: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N — yes		
Battery box	- New box with new batteries: Y/N - Old box with new batteries: Y/N - Moisture inside: Y/N - New desiccant bags (2): Y/N !		
Stakes	- New ones drilled in: Y/N - Length: 6 / 8 / 10 m - Caps in bottom end to reduce melt-in: Y/N		
Tripod	- New parts: Y/N - Boom direction: true north-south: Y/N fixed a misplaced eyebol to level station :)		

Metadata after maintenance / establishment

Time logger clock set (to UTC)?	Y / N
Name of logger program	2014 A
Photo or screen dump of values in fast scan mode	Y / N
Latitude (dd mm.mmm)	
Longitude (dd mm.mmm)	
Photos of tripod, sensors, logger box wiring etc. (more is better)	Y / N
Mast tilt in boom direction (+ if radiometer tilting down)	-0.8 °
Mast tilt across boom looking from radiometer (- if clockwise)	+0.3 °
Boom direction relative to north (radiometer should point to true south)	° true/magn
Box on wind sensor exactly towards mast? If not, change.	☒ Y / N
Radiometer aligned with mast and inclinometer? If not, align.	☒ Y / N
Temperature / humidity height (bottom of casing – surface)	cm
Sonic ranger on AWS height (membrane – surface)	cm
Wind sensor height (center of propeller – surface)	cm
Sonic ranger on stakes height (membrane – surface)	cm
Free length of stake – outer, holding boom with sonic ranger	cm
Free length of stake – middle	cm
Free length of stake – outer, holding boom without sonic ranger	cm
Free length of ablation hose on/above the ice surface	cm
Alt: depth of ablation hose using markings on hose	cm
Height of ablation hose fluid level above the ice surface	cm
Vertical difference in surface height (at station – at hose) (approx)	cm
Length of thermistor string on surface from surface marking	cm
Height of surface irregularities	cm

Notes: